

Research Notes

Network Scanning & Service Enumeration

Introduction

Network scanning is an important phase of ethical hacking that helps identify active hosts, open ports, running services, and operating systems on a network. Security professionals use network scanning to understand a system's attack surface before performing a vulnerability assessment or penetration test. Scanning should only be performed on systems where permission has been granted.

What is Nmap?

Nmap (Network Mapper) is a free and open-source network scanning tool used by ethical hackers, network administrators, and security professionals. It is designed to discover hosts and services on a network by sending packets and analyzing the responses.

Main Features of Nmap

- Host discovery
- Port scanning
- Service version detection
- Operating system detection
- Network inventory
- Security auditing
- Vulnerability assessment support

Common Nmap Commands

Command	Purpose
nmap localhost	Scan the local computer
nmap 127.0.0.1	Scan the local IP address
nmap -sV localhost	Detect service versions
nmap -O localhost	Detect the operating system
nmap -A localhost	Perform an advanced scan

Network Scanning

Network scanning is the process of examining a system or network to identify:

- Active devices
- Open ports
- Running services
- Network configuration
- Potential security weaknesses

It is one of the first steps in ethical hacking because it provides information needed for further security testing.

TCP vs UDP

TCP (Transmission Control Protocol)

- Connection-oriented protocol
- Reliable data transmission
- Performs error checking
- Used for SSH, HTTP, HTTPS, FTP

UDP (User Datagram Protocol)

- Connectionless protocol
- Faster than TCP
- No guaranteed delivery
- Used for DNS, VoIP, video streaming

Service Enumeration

Service enumeration is the process of collecting detailed information about services running on open ports. It identifies the service name, version, and other useful details that help security professionals determine whether the service has known vulnerabilities.

Example:

- Port 22 → SSH
- Port 80 → HTTP
- Port 443 → HTTPS

Banner Grabbing

Banner grabbing is a technique used to collect information about a network service. The information may include:

- Software name
- Software version
- Operating system
- Server type

This helps identify outdated software that may contain security vulnerabilities.

Operating System Fingerprinting

Operating system fingerprinting is a method used to determine the operating system running on a target computer. Nmap analyzes responses to specially crafted packets and compares them with its database to identify the operating system.

Knowing the operating system helps security professionals:

- Select appropriate testing techniques.
- Identify operating system–specific vulnerabilities.
- Improve the accuracy of security assessments.

Common Network Ports

Port	Service	Purpose
20/21	FTP	File transfer
22	SSH	Secure remote login
23	Telnet	Remote login (unencrypted)
25	SMTP	Email sending
53	DNS	Domain name resolution
80	HTTP	Web browsing
110	POP3	Receiving emails
143	IMAP	Email synchronization
443	HTTPS	Secure web browsing

445	SMB	File and printer sharing
3389	RDP	Remote Desktop access

Importance of Network Scanning

Network scanning helps organizations:

- Discover active devices.
- Identify open ports.
- Detect unnecessary services.
- Reduce the attack surface.
- Improve network security.
- Support vulnerability assessments and penetration testing.

Security Best Practices

- Scan only systems you own or have permission to test.
- Keep operating systems and software updated.
- Disable unused services and ports.
- Use strong passwords and multi-factor authentication.
- Configure firewalls to restrict unnecessary access.
- Perform regular security scans using Nmap.
- Monitor systems for unusual network activity.

Key Learning Outcomes

Through this task, I learned how to use Nmap to perform network scanning and service enumeration on my local machine. I understood how to identify open ports, detect running services, and perform operating system detection. I also learned the importance of securing network services, closing unnecessary ports, and following ethical guidelines when conducting security assessments. This exercise improved my practical knowledge of network scanning and demonstrated how it supports vulnerability assessment and penetration testing while helping organizations strengthen their cybersecurity.